

CHAPTER - 4

IDENTIFYING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The Lenskart E-Commerce Database Management System is designed to manage the sale of eyeglasses, sunglasses, and contact lenses along with prescription-based ordering. It handles customers, categories, brands, products, prescriptions, carts, orders, order items, payments, reviews, deliveries, and stores. Functional dependencies help identify how non-key attributes relate to the primary key in each table, while redundancy analysis and data anomaly detection ensure that data remains consistent, accurate, and free of unnecessary duplication. A well-designed database of this kind improves query performance, maintains referential integrity, and supports reliable order processing and prescription tracking for an eyewear e-commerce platform.

4.2 CUSTOMER TABLE

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_No, Date_of_Birth, Address
--

Redundancy Analysis

Customer details are stored only once in the Customer table. Cart, Order, Prescription, and Review reference the customer using Customer_ID instead of repeating customer information.

Data Anomalies

- Insert Anomaly: A new customer can be registered even before placing an order or adding a prescription.
- Update Anomaly: Updating a customer's address or phone number requires a change only in the Customer table.
- Delete Anomaly: Deleting an order does not remove the customer's record.

Attribute	Key Type
Customer_ID	Primary Key
Customer Name	Not Null
Email	Unique Key

Phone No	Unique Key
Address	-

4.3 CATEGORY TABLE

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Each category is stored only once. Products reference the category using Category_ID instead of repeating the category name and description.

Data Anomalies

- Insert Anomaly: A new category can be added without any product being listed under it.
- Update Anomaly: Renaming a category or editing its description is done only once.
- Delete Anomaly: Deleting a product does not delete the category.

Attribute	Key Type
Category_ID	Primary Key
Category Name	Unique Key
Description	-

4.4 BRAND TABLE

Functional Dependency

Brand_ID → Brand_Name, Contact_No, Address

Redundancy Analysis

Brand details are stored only once in the Brand table. Products reference the brand using Brand_ID instead of repeating brand information.

Data Anomalies

- Insert Anomaly: A new brand can be added even before listing any product.
- Update Anomaly: Updating a brand's contact number or address requires a change only in the Brand table.
- Delete Anomaly: Deleting a product does not remove the brand's record.

Attribute	Key Type
Brand_ID	Primary Key
Brand Name	Not Null
Contact_No	Unique Key
Address	-

4.5 PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Category_ID, Brand_ID, Frame_Type, Price, Stock, Power-Compatible

Redundancy Analysis

Product details are stored only in the Product table. Order_Item and Review records use Product_ID instead of repeating product name, price, or frame details for every order or review.

Data Anomalies

- Insert Anomaly: A new product can be listed by a brand before it receives any order.
- Update Anomaly: Changing a product's price or stock is updated only once in the Product table.
- Delete Anomaly: Deleting an Order_Item does not remove the product listing itself.

Attribute	Key Type
Product_ID	Primary Key

Product Name	-
Category_ID	Foreign Key
Brand_ID	Foreign Key
Frame_Type	-
Price	-
Stock	-
Power_Compatible	-

4.6 PRESCRIPTION TABLE

Functional Dependency

Prescription_ID → Customer_ID, Right_Eye_Power, Left_Eye_Power, Cylindrical_Power, Axis, Prescription_Date

Redundancy Analysis

Prescription details are stored separately from the Customer table, and Order_Item references Prescription_ID instead of repeating eye power values for every order.

Data Anomalies

- Insert Anomaly: A prescription can be saved by a customer before it is ever applied to an order.
- Update Anomaly: Updating eye power values affects only that specific prescription record.
- Delete Anomaly: Deleting an Order_Item does not remove the customer's saved prescription record.

Attribute	Key Type
Prescription_ID	Primary Key
Customer_ID	Foreign Key
Right_Eye_Power	-
Left_Eye_Power	-
Cylindrical_Power	-
Axis	-

Prescription_Date	-
-------------------	---

4.7 CART TABLE

Functional Dependency

Cart_ID → Customer_ID, Total_items, Total_amount
--

Redundancy Analysis

The Cart table stores only cart-level information. Customer details are referenced using the Customer_ID foreign key rather than being repeated.

Data Anomalies

- Insert Anomaly: A cart is created for a customer even before any item is added to it.
- Update Anomaly: Updating Total_items or Total_amount does not affect the customer's stored details.
- Delete Anomaly: Deleting a cart does not remove the customer's record.

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_items	-
Total_amount	-

4.8 ORDER TABLE

Functional Dependency

Order_ID → Customer_ID, Order_date, Order_status, Total_amount
--

Redundancy Analysis

The Order table stores only order-level information. Customer details are referenced using the Customer_ID foreign key rather than being repeated.

Data Anomalies

- Insert Anomaly: An order can only be created once a Customer_ID already exists, since every order must belong to a registered customer.
- Update Anomaly: Order_status can be updated (e.g., Pending to Shipped) without affecting customer or product data.
- Delete Anomaly: Deleting an order does not remove the customer record, but does affect its linked Order_Items, Payment, and Delivery if cascading is not handled carefully.

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_date	-
Order_status	Check Key
Total_amount	-

4.9 ORDER_ITEM TABLE

Functional Dependency

Order_item_ID → Order_ID, Product_ID, Prescription_ID, Quantity, Price

Redundancy Analysis

This table connects Orders, Products, and Prescriptions, avoiding repeated product or prescription information inside the Order table itself.

Data Anomalies

- Insert Anomaly: A product can be added as an order line item only if the Order and Product already exist, and the Prescription if it is a power-compatible item.
- Update Anomaly: Changing the Quantity or Price of one item does not affect any other order item.
- Delete Anomaly: Removing an order item does not delete the underlying

product or prescription record.

Attribute	Key Type
Order_item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Prescription_ID	Foreign Key
Quantity	-
Price	-

4.10 PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_method, Payment_status, Payment_date, Amount

Redundancy Analysis

Payment information is stored separately from the Order table, avoiding duplicate payment records for the same order (the One-to-One relationship ensures a single payment entry per order).

Data Anomalies

- Insert Anomaly: A payment record can be created only after the corresponding order exists.
- Update Anomaly: Payment_status can be updated (e.g., Pending to Paid) without affecting the order or customer data.
- Delete Anomaly: Deleting a payment record does not remove the associated order, but leaves the order without proof of payment.

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_method	-
Payment_status	Check Key

Payment_date	-
--------------	---

4.11 REVIEW TABLE

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_text, Review_date

Redundancy Analysis

Review information is stored separately from Customer and Product tables, avoiding repetition of customer or product details for every review submitted.

Data Anomalies

- Insert Anomaly: A review can be added only if both the Customer and Product already exist in their respective tables.
- Update Anomaly: Editing a review's rating or text affects only that specific review record.
- Delete Anomaly: Deleting a review does not remove the customer's or product's record.

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check Key
Review_text	-
Review_date	-

4.12 DELIVERY TABLE

Functional Dependency

Delivery_ID → Order_ID, Delivery_address, Delivery_date, Delivery_status
--

Redundancy Analysis

Delivery information is stored separately from the Order table (the One-to-One relationship ensures a single delivery entry per order), avoiding repetition of order-level data.

Data Anomalies

- Insert Anomaly: A delivery record can be created only after the corresponding order exists.
- Update Anomaly: Delivery_status can be updated (e.g., Dispatched to Delivered) without affecting the order or payment data.
- Delete Anomaly: Deleting a delivery record does not remove the associated order, but leaves the order without tracking information.

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_address	-
Delivery_date	-
Delivery_status	Check Key

4.13 STORE TABLE

Functional Dependency

Store_ID → Store_Name, Location, Contact_No

Redundancy Analysis

Store details are stored only once in the Store table. As Store is not referenced as a foreign key by any other entity in this schema, it functions as an independent record for offline pickup or franchise information.

Data Anomalies

- Insert Anomaly: A new store can be added independently, without any product,

order, or customer data.

- Update Anomaly: Updating a store's location or contact number requires a change only in the Store table.
- Delete Anomaly: Deleting a store record does not affect any customer, order, or product data, since no other table currently references Store_ID.

• Attribute	Key Type
Store_ID	Primary Key
Store_Name	-
Location	-
Contact_No	Unique Key

4.14 CONCLUSION

The Lenskart E-Commerce Database Management System is designed using proper functional dependencies and normalized tables for Customers, Categories, Brands, Products, Prescriptions, Carts, Orders, Order_Items, Payments, Reviews, Deliveries, and Stores. Each entity stores only its relevant data, with foreign keys used to link related information instead of repeating it, thereby reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains referential integrity, and ensures an efficient and reliable database for managing eyewear catalogues, prescription-based ordering, and order fulfilment on an e-commerce platform.